Daily Drilling Report
SPD19B-03
9/10/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
45.0	8 1/2"	190.00	72.83	31.70	103.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	264,057.41	2,450.38	11,049,073.01		40.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
3,039.00	3,232.00		193.00	13.85	13.9

Operations at Report Time
Sweeping the hole clean.

Operations Summary
Directional drilling 8 1/2" hole section f/ 3039m to 3232m TD. Sweep the hole. Perform wiper trip f/ 3232m to 2950m.

Operations next 24hrs
Cont' wiper trip to 9 5/8" CSG shoe. RIH back to BTM. Sweep hole. Pump out of the hole to 400m inside 9 5/8" CSG. Normal POOH drilling BHA#11 to surface. Perform wireline logging job.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Behnam Jahangiri	Night Drilling Sup.
Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	9/4/2014	9/26/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	9/6/2014	10/7/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/10/2014	10/26/2014
Trip Drill	9/8/2014	10/29/2014

STOP Cards

Number of Reports: 34

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	19:00	19....	8 1/2"	D	03	P	Cont' directional drilling 8 1/2" hole in rotation mode f/ 3039m to 3332m TD w/ 550 gpm/ 2450-2600 psi/ 17-25 kips/ 60 rpm/ 18-25 TRQ. * Drilling with NL/NG. * Pump 20 bbl Hi-Vis pill mid-stand & 20 bbl when STD down. * B/R and R/D each STD drilled once. * Take survey before connection.
19:00	20:30	1.50	8 1/2"	D	07	P	Sweep hole by pumping 20 bbl Lo-Vis + 20 bbl Hi-Vis. Circulate w/ 550 gpm / 2250 psi / 40 rpm. Meanwhile reciprocate string. Confirm the pill is out of hole and shakers clean.
20:30	00:00	3.50	8 1/2"	D	06	P	Perform wiper trip f/ 3232m to 2950m w/ 450 gpm / 1550 psi / 40 rpm / 13-17 Kft-lbs.
00:00	00:00						Sep 11, 2014 00:00-02:30 Cont' B/Reaming f/ 2950m to 2720m (9 5/8" Shoe) w/ 450 gpm / 1550 psi / 40 rpm / 13-17 Kft-lbs. 02:30-05:00 Flow check inside shoe, well static. RIH on elevator f/ 2720m to bottom (3232m), good hole condition. 05:00-06:00 Sweep the hole by pumping 20 bbl Lo-Vis + 20 bbl Hi-Vis tandem pill. Circulate w/ 550 gpm / 2250 psi / 15 rpm. Meanwhile reciprocate the string. >> Last sample @ 3232m: Lst. (10 %) + Anhy. (90%)

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
3,232.00	CaCo3/KCL/ Polymer	10.50	55	24.0	21.000	2.8
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.5	0.500	1.400	1.200			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	7	0.0	87.0	13.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
6.000	8.000	11.000	0/100	61.0	1438.0	2605.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
7589.0	136.0	0.0	1.34		13.39637	0.396

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Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
2.00	7.	335	18.0	999.0	36.6

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
EP-HUA	Chopper			Pax in:8 Pax out:0
EP-HTN	Chopper			Pax in:3 Pax out:0
EP-HUA	Chopper			Pax in:7 Pax out:6
EP-HTN	Chopper			Pax in:7 Pax out:6
EP-HBJ	Chopper			Pax in:7 Pax out:8
Ahmed 6	Supply Boat	9/7/2014 17:30		C/out chopper safety watch.Streaming at location due to bad WX.
Marina 1	Supply Boat			Standby at Sahar-2 Anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
CaCo3-F BB	4.0	174.0	59.0
Caustic Soda 2014	4.0	500.0	283.0
Duo-Tec	7.0	185.0	103.0
MI-Cide	4.0	47.0	31.0
Starch	2.0	1,600.0	355.0
Pac-LV	4.0	521.0	69.0
Soda Ash 25 Kg	4.0	750.0	268.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	25.0	18.0	168.0
Cement G	0.0	0.0	92.0
Drill Water	0.0	0.0	304.0
Diesel Fuel	9.0	0.0	145.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	129.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
11	8 1/2	Varel	VM713GUX	6006868	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	23.45	299,566.80	2,949.00	3,232.00	283.00	12.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
500	550	2,150.0	2,600.0	5	25	170	214
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
33.90	23.45	100.00	0.00	470	378	301	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
3,039.00	3,210.50	11.35	21,000.0	550	60	2,600.0	20	378
3,210.50	3,232.00	2.50	21,000.0	550	60	2,500.0	25	378

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWPD	5 7/8	4.000	27.89	252.82
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	224.93
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	219.13
X/O	6 1/2	2.813	0.85	32.78
6 3/4" Float sub	6 3/4		0.81	31.93
6 3/4" NMDC	6 3/4	3.250	9.48	31.12
6 3/4" TM - Sub	6 3/4		1.74	21.64
6 3/4" NMDC	6 3/4	3.250	9.48	19.90
8" St. Stabilizer (COR-2941AD)	8	3.000	2.14	10.42
6 3/4" Motor (AKO: 1.25° , Sleeve: 8 3/8")	6 3/4		8.00	8.28
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
2	3,180.00	14.00	20	295.0	97	114
2	3,180.00	14.00	30	450.0	97	170
2	3,180.00	14.00	40	620.0	97	227
3	3,180.00	14.00	20	300.0	97	114
3	3,180.00	14.00	30	445.0	97	170
3	3,180.00	14.00	40	620.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

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Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
3,048.15	249.80	26.40	2,872.74	-385.49	-876.66	957.67	0.29	0.10
3,077.42	249.60	26.40	2,898.95	-390.00	-888.87	970.66	0.09	0.00
3,106.10	249.10	26.50	2,924.63	-394.51	-900.82	983.42	0.26	0.10
3,135.50	248.80	26.60	2,950.93	-399.23	-913.09	996.55	0.17	0.10
3,164.32	248.60	26.60	2,976.70	-403.91	-925.11	1,009.44	0.09	0.00
3,193.05	248.10	26.80	3,002.37	-408.68	-937.11	1,022.35	0.31	0.21
3,213.73	247.80	26.80	3,020.83	-412.18	-945.75	1,031.67	0.20	0.00

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
K1		2,732.00	2,586.77
K2		2,851.80	2,696.55
K3		2,901.00	2,740.86
K4		3,029.50	2,856.03
Nar		3,198.50	3,007.23

General Remarks

Comment

>> Requirements: (200 sxs Starch f/ MR #14)

>>> Sent:

** Nil

>>> Received:

** Nil